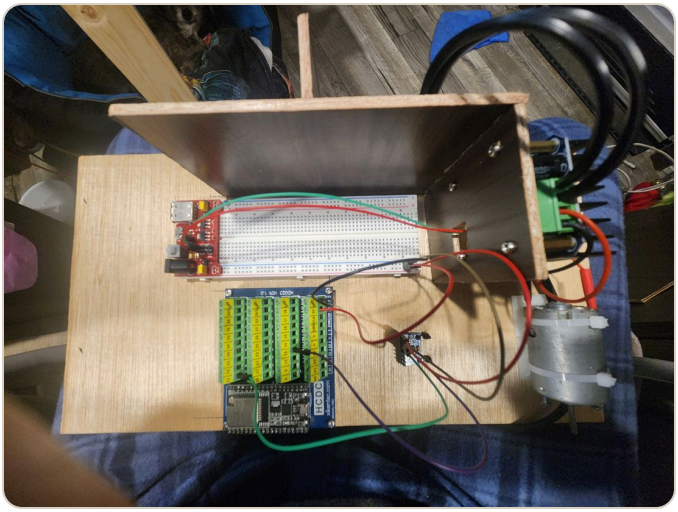
Awning Controller Test Rig

The bench mock-up on plywood, October 2026. It runs the real awning firmware, so everything tested here carries over to the box on the arm.



Your rig on 26 October 2026, before the motor driver was wired. The motor now runs both ways from Home Assistant.

What's on the board

- **ESP32** on the HCDC screw-terminal board, powered by USB
- **GY-521** (MPU-6050): arm tilt and wind shake. **Working.**
- **BTS7960** motor driver, under the plywood shelf
- **Small test motor** standing in for the awning motor
- **Breadboard** with power module: holds the two 10k pull-downs and ties the grounds together
- Also mounted but **not used by the awning firmware**: microphone, PIR motion sensor, microswitch, buck converter (for later)

Different from the final box

- **No current sensor yet:** a jumper ties VN to GND so the firmware reads 0 A
- **No diodes and no wall rocker:** the diodes only matter once the rocker is connected
- **Breadboard instead of perfboard**
- **No reed switch yet**, so the firmware never sees "fully closed"

1. Test rig wiring

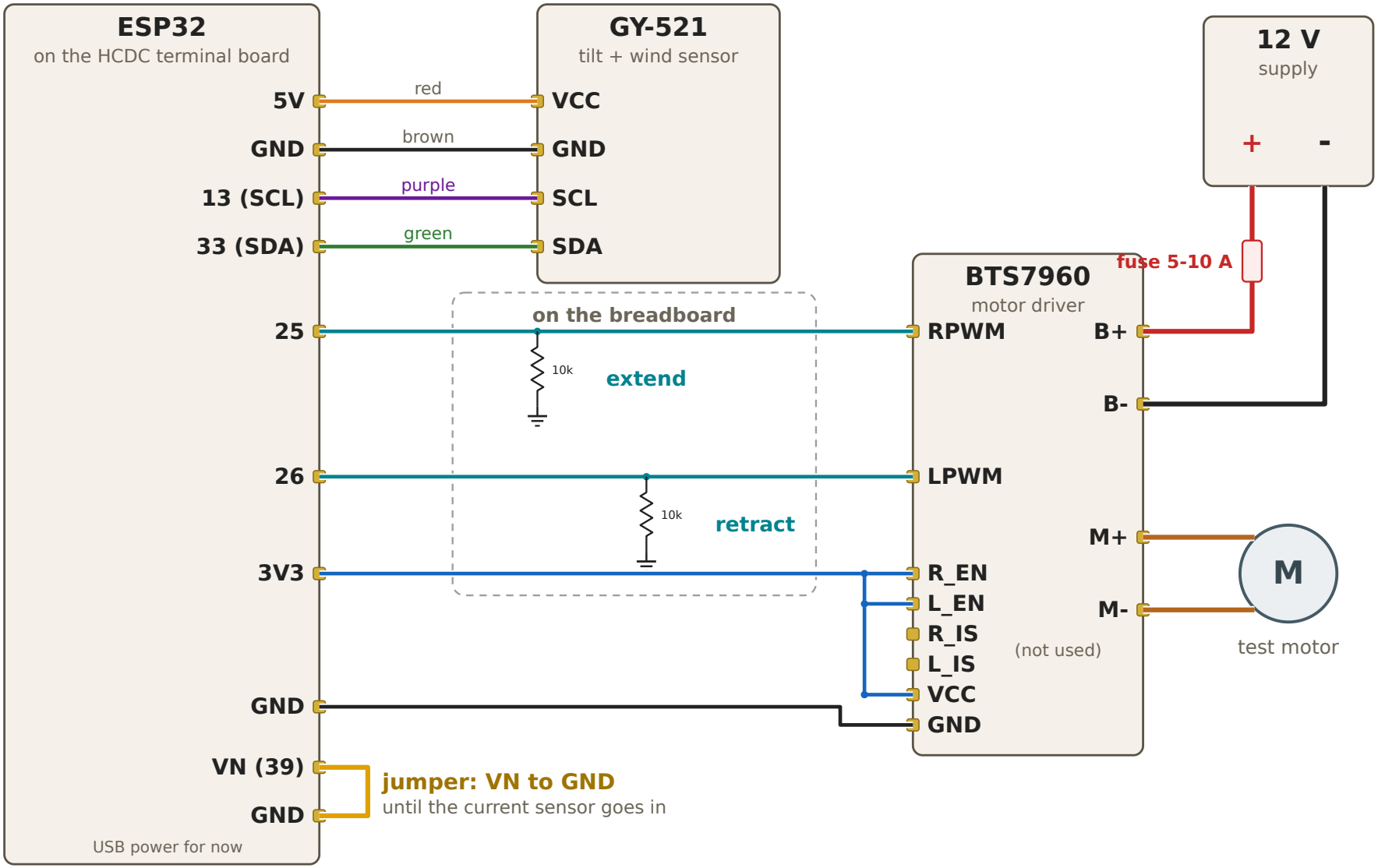


Figure 1: Everything on the plywood. Ground symbols are the breadboard's blue (-) rail, which is tied to ESP32 GND. Colors match the wires you used where they're already in. The ESP32 pins are grouped by job, not in their order on the terminal board.

2. Wiring checklist

Work down the list with the 12 V supply **unplugged**. On the screw terminals, use stripped wire or male pins, never a female Dupont socket.

	From	To	Wire
✔ done	**ESP32 5V**	GY-521 VCC	red
✔ done	**ESP32 GND**	GY-521 GND	brown
✔ done	**ESP32 13**	GY-521 SCL	purple
✔ done	**ESP32 33**	GY-521 SDA	green
✔ done	**ESP32 GND**	breadboard blue (-) rail	black
✔ done	**ESP32 VN**	ESP32 GND (jumper)	any
✔ done	**ESP32 25**	breadboard row A, then BTS7960 RPWM	
✔ done	**10k resistor**	row A to the blue (-) rail	
✔ done	**ESP32 26**	breadboard row B, then BTS7960 LPWM	
✔ done	**10k resistor**	row B to the blue (-) rail	
✔ done	**ESP32 3V3**	BTS7960 VCC, R_EN and L_EN (all three)	
✔ done	**ESP32 GND**	BTS7960 GND	
✔ done	**12 V + (via fuse)**	BTS7960 B+	
✔ done	**12 V -**	BTS7960 B-	
✔ done	**test motor**	BTS7960 M+ and M-	

3V3, not 5V, goes to the BTS7960's VCC, R_EN and L_EN. Never connect the 12 V supply to the small header pins. 12 V goes only to the big B+ / B- screws.

3. Meter check (before the motor runs)

1. ESP32 on USB, 12 V **off**. Meter on DC volts.
2. **RPWM to GND** and **LPWM to GND**: both about **0 V**.
3. **VCC to GND** on the BTS7960: about **3.3 V**.
4. Plug in the 12 V. **M+ to B-** and **M- to B-**: both **0 V**, and the motor does **not** move.
5. Unplug and replug the ESP32's USB: the motor must stay still during the restart.

4. First test run

1. Open the **Awning** page in Home Assistant. The device is called **Camper Awning**.
2. Check **Arm Tilt** changes when you tip the plywood, and **Motor Current** reads 0.0 A.
3. Press **open** for a second or two, then **stop**. The motor should spin one way.
4. Press **close** for a second or two, then **stop**. It should spin the other way.
5. If **open** spins the wrong way, swap the two motor wires on M+ and M-.
6. Leave **Wind Auto-Retract off** on the bench: tapping the plywood looks like wind.

> **On the bench, the firmware can't tell where the awning is.** Arm Tilt only means something once the sensor is on the arm and calibrated. A run may stop on its own after a few seconds ("no progress") or at the 45-second safety limit. That's the firmware protecting the awning, not a fault.

5. Later, for the real box

- Remove the VN jumper and add the ACS712 with its 10k/10k divider (guide, page 10).
- Move the pull-downs to the perfboard and add the two diodes, stripe toward RPWM / LPWM. Use 1N5817; the 1N4001s work but leave little margin.
- Add the reed switch (GPIO32 to GND), the wall rocker wires, and the buck converter for 12 V to 5 V.